MILITARY STANDARD

TRUCKLOADING OF HAZARDOUS MATERIALS

BOOSTER MK 11 MODS 2 & 5 (TALOS)

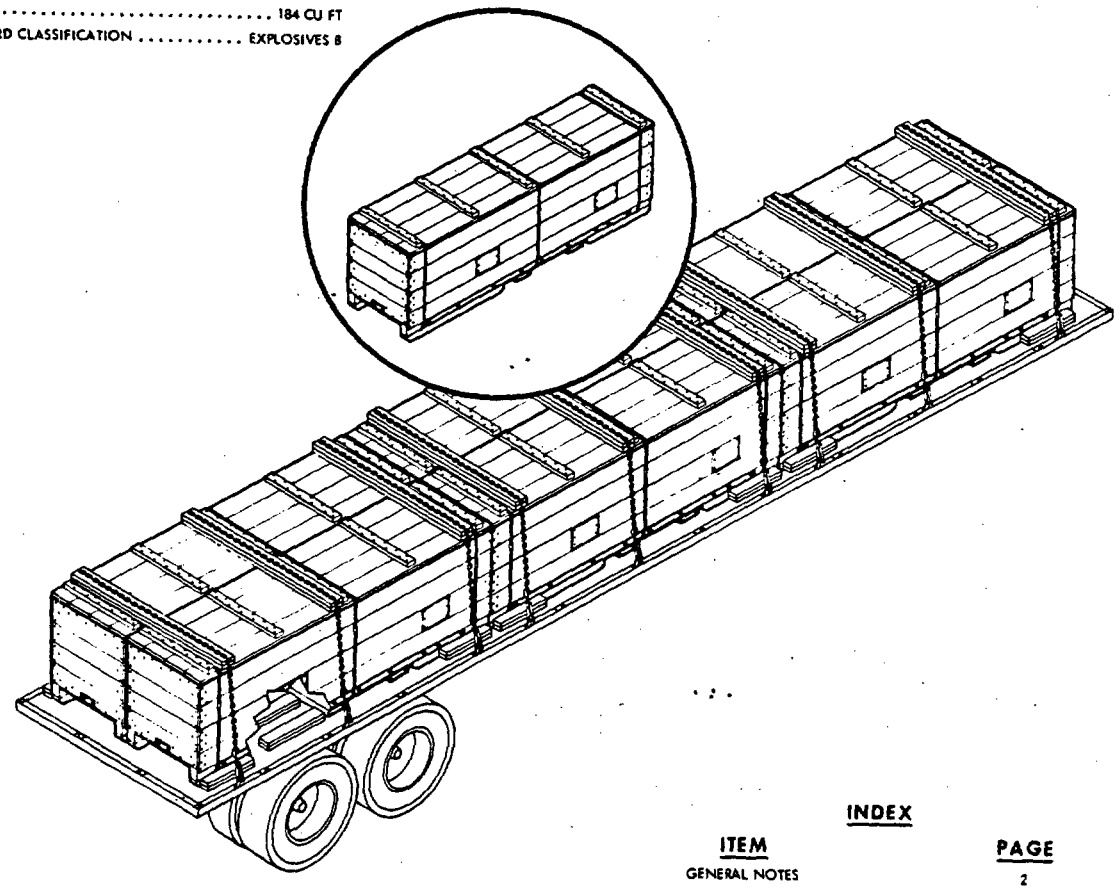
IN CONTAINER MK 576 MOD 0

MIL-STD-1320-101
(NAVY)

9 JUNE 1975

CONTAINER DATA

DIMENSIONS	155 L x 40 W x 51 1/4 H
GROSS WT. (APPROX)	5435 LBS
NET WT. (APPROX)	4220 LBS
TARE WT. (APPROX)	1215 LBS
CUBE	184 CU FT
DOT HAZARD CLASSIFICATION	EXPLOSIVES B



NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINER-LOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

INDEX

ITEM	PAGE
GENERAL NOTES	2
FTL 40 FT FLATBED	2 & 3
LTL 40 FT FLATBED	4 & 5
LOAD STRAPPING	4

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M S Gray NWHL 6/2/75
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

A E Kelly 6/2/75
SIGNATURE SEASYSCOM BY DIRECTION DATE

ORIGINATOR
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

Source: https://assist.dla.mil -- Downloaded: 20260-23T23:43Z
Check the source to verify that this is the current version before use.

GENERAL NOTES

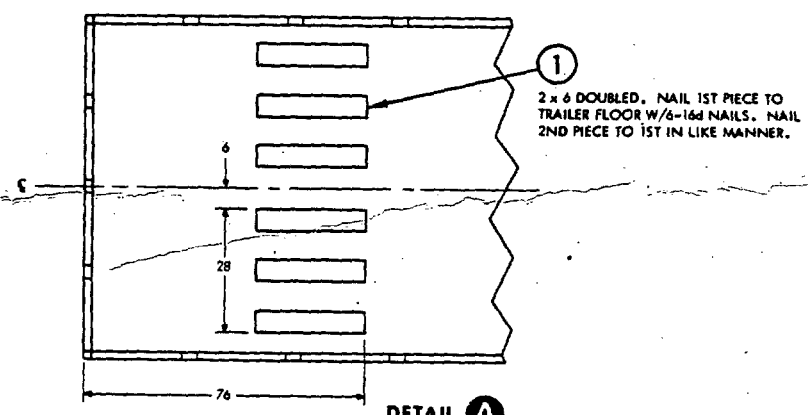
1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE BOOSTER MK 11 MODS 2 AND 5 IN CONTAINER MK 576 MOD 0.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT FLATBED TRAILERS WITH THE AXLES LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS.
3. TRAILERS MAY HAVE EXPOSED STEEL STRUCTURE MEMBERS RUNNING FORE AND AFT. LATERAL LOCATION OF BACKUP CLEATS, PIECE 1, MAY HAVE TO BE VARIED. DO NOT VARY FORE AND AFT LOCATION OF BACKUP CLEATS.
4. CHAINS AND STEEL STRAPS MAY BE USED INTERCHANGEABLY FOR TIE-DOWNS ON A 1 TO 1 BASIS.
5. STEEL STRAPS SHALL BE 2 x .050 AND APPLIED AS SPECIFIED BY THE LOAD STRAPPING REQUIREMENTS SHOWN ON PAGE 4.
6. WHEN USING CHAIN OR STRAPS, INSERT A PROTECTOR BOARD (LAMINATED DOUBLE 2 x 6) BETWEEN CHAIN AND LADING. SECURE CHAIN TO THE BOARD WITH FOUR 20d NAILS BY DRIVING THE NAIL THROUGH THE CHAIN INTO THE BOARD AND CLINCHING IT OVER THE CHAIN.
7. CHAINS AND LOAD BINDERS SHALL MEET THE FOLLOWING CONDITIONS:
   a. CHAINS AND FITTINGS SHALL BE 3/8 INCH AND LOADBINDERS 3/8 INCH CAPACITY.
   b. CHAINS, FITTINGS AND LOADBINDERS SHALL HAVE A MINIMUM SAFE WORKING LOAD OF 5000 LBS.
   c. LOADBINDERS SHALL BE SAFETY WIRED WITH 16 GAUGE SOFT ANNEALED IRON WIRE OR SECURED USING THE END PIECE OF THE 3/8 CHAIN.
8. PRIOR TO LOADING THE TRAILER AND DURING THE PRELOADING INSPECTION REQUIRED BY OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOADBINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOADBINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOADBINDER.
9. THE LADING SHALL BE COVERED WITH A WATERPROOF, FIRE RESISTANT TARPAULIN. THE BEST RESULTS ARE OBTAINED IF THE CONTAINERS ARE COVERED WITH THE TARPAULIN PRIOR TO APPLYING THE TIE-DOWN CHAIN OR STRAPS. PROTECTOR BOARDS SHALL BE USED BETWEEN THE TIE-DOWNS AND TARPAULINS.
10. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO TRAILER IN AN ACCESSIBLE AREA AND ATTACH "EXPLOSIVE B" PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER.
12. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II STYLE 10, COMMON BRIGHT; STRAPPING, QQ-S-781, TYPE I CLASS A.
13. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL 40 FT TRAILER (FLATBED)

1. THE TRAILER SHALL BE LOADED STARTING AT THE REAR AS FOLLOWS:
   a. INSTALL PREPOSITION BLOCKING AS SHOWN IN DETAIL.
   b. POSITION REAR STACK OF TWO CONTAINERS ON THE TRAILER WITH THE CONTAINER FORK POCKET AGAINST THE CLEATS SHOWN ON PAGE 3.
   c. POSITION CENTER STACK OF CONTAINERS ON TRAILER.
   d. MEASURE 110 3/4 INCHES FROM FORWARD END OF CONTAINERS IN CENTER STACK AND INSTALL PREPOSITION BLOCKING (SEE DETAIL FOR CLEAT SPACING).
   e. POSITION FORWARD STACK ON TRAILER AND COMPLETE REMAINING BLOCKING AND TIE-DOWNS AS SHOWN ON PAGE 3.

WARNING

TO ENGAGE THE PREPOSITION BLOCKING, CONTAINERS SHALL BE POSITIONED WITH THE CLAMP ASSEMBLY END TO THE REAR OF THE TRAILER.



DETAIL A

REAR PREPOSITIONED BLOCKING

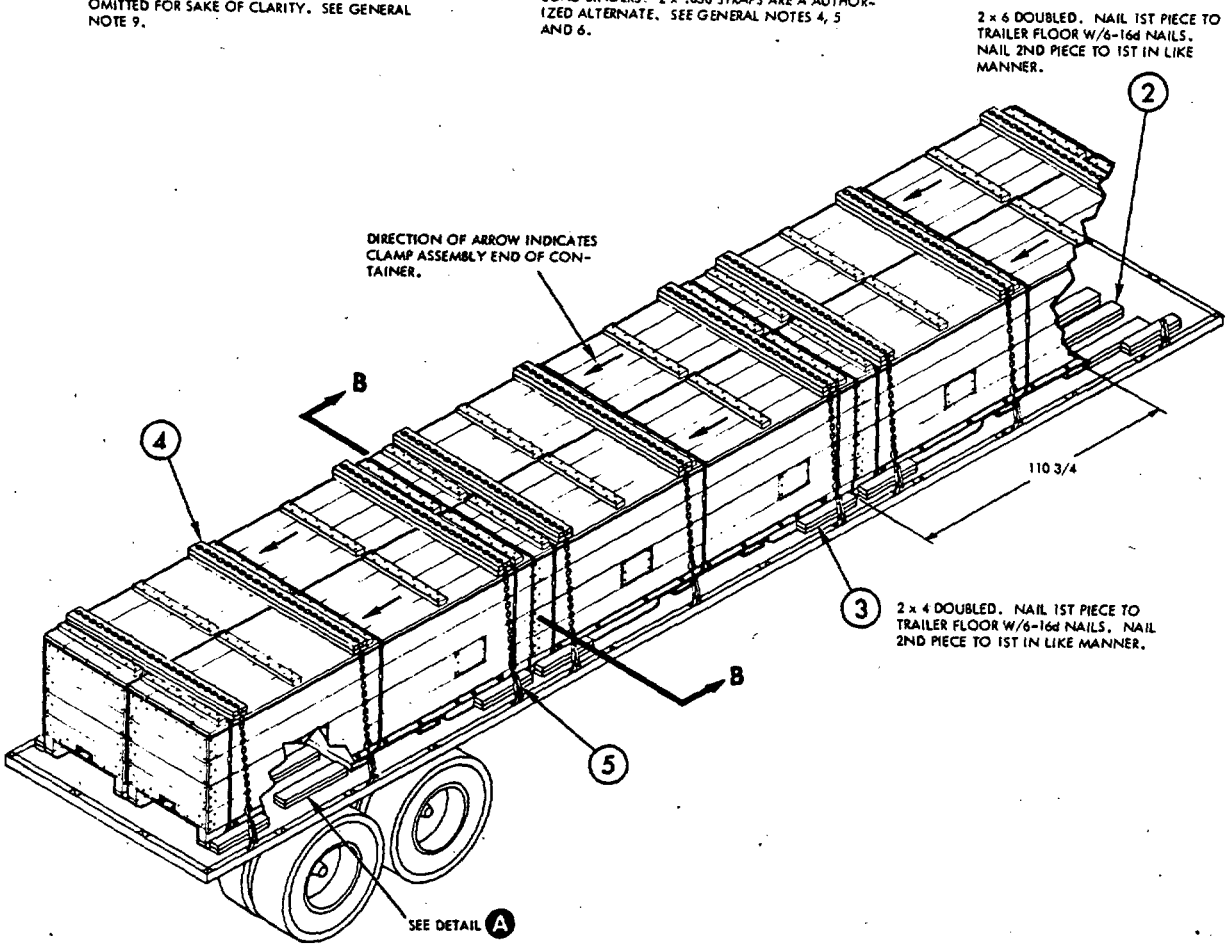
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:43Z
Check the source to verify that this is the current version before use.

NOTE

THIS LOAD SHALL BE COVERED WITH A TARPAULIN. IN THE LOAD BELOW, THE TARPAULIN HAS BEEN OMITTED FOR SAKE OF CLARITY. SEE GENERAL NOTE 9.

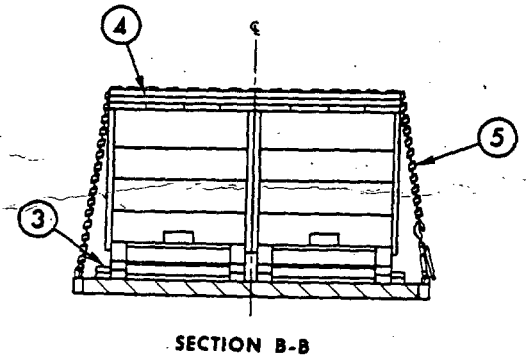
NOTE

THIS FTL ILLUSTRATES THE USE OF CHAINS AND LOAD BINDERS. 2 x .050 STRAPS ARE A AUTHORIZED ALTERNATE. SEE GENERAL NOTES 4, 5 AND 6.



TRUCKLOAD DATA

NO. OF CONTAINERS 6
WEIGHT (APPROX) . . . . . 32,610 LBS



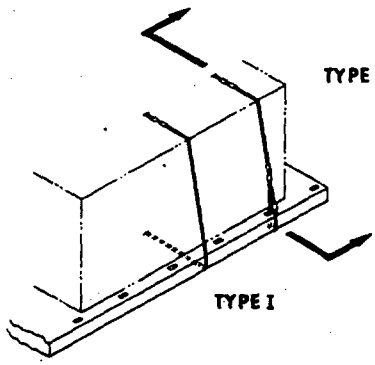
SECTION B-B

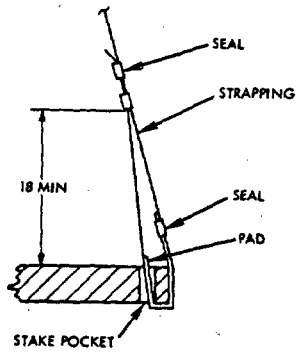
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
5	CHAIN & LOADBINDER	3/8	9	-	-	-
4	PROTECTOR BOARD	2 x 6 x 80	18	DOUBLE LAMINATE	8	8d
3	SLEEPER	2 x 4 x 24	24	SEE FIELD NOTE		
2	FWD BACKUP CLEAT	2 x 6 x 30	12	SEE FIELD NOTE		
1	AFT BACKUP CLEAT	2 x 6 x 30	12	SEE DETAIL A		

LIST OF MATERIALS AND NAILING DATA

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:43Z
Check the source to verify that this is the current version before use.

LOAD STRAPPING





SECTION C-C

TIE-DOWN STRAPPING

THE FOLLOWING TWO METHODS OF TIE-DOWN STRAPPING ARE APPROVED FOR USE WITH THE SHIPMENT OF LOADS ON FLATBED TRAILERS. STEEL STRAPPING SHALL BE IN ACCORDANCE WITH FEDERAL SPECIFICATION QQ-S-781. STRAPPING SEALS SHALL HAVE A MINIMUM TENSILE STRENGTH EQUAL TO 75 PER CENT OF THE STRAP STRENGTH. FOR EACH OF THE TYPES ILLUSTRATED ABOVE IT IS PREFERRED TO POSITION, TENSION AND DOUBLE CRIMP THE STRAP SEALS AT THE TOP OF THE LOAD, IF PRACTICABLE.

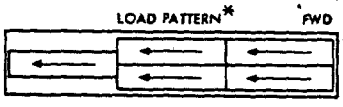
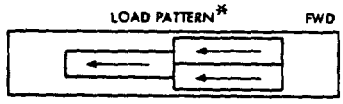
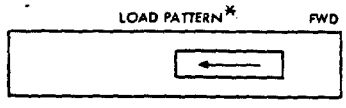
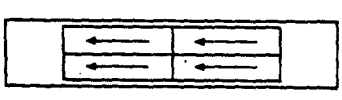
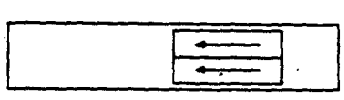
TYPE I - CONTINUOUS STRAPPING AROUND THE FLATBED AND THE LOAD.

TYPE II - THE STRAPPING IS SECURED TO THE STAKE POCKETS, ONE PIECE ON EACH SIDE OF THE TRAILER, AND IS BROUGHT UP OVER THE LOAD, TENSIONED, AND SEALED WITH TWO DOUBLE-CRIMPED SEALS ON THE TOP. METHOD OF SECURING STRAPPING TO STAKE POCKET IS SHOWN IN SECTION C-C. THE SHORT END IS ON THE INSIDE AND IS SECURED WITH TWO DOUBLE-CRIMPED SEALS AT A MINIMUM OF 18 INCHES ABOVE THE TRAILER BED. A STAKE POCKET PAD (A SHORT PIECE OF THE SAME STRAPPING 18 INCHES LONG) IS INSERTED BETWEEN THE MAIN STRAP AND THE STAKE POCKET AND IS SECURED TO THE MAIN STRAP WITH A SEAL AS SHOWN.

LTL 40 FT TRAILER (FLATBED)

1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD (LTL), SELECT FROM TABLE I, THE CORRECT LOAD PATTERN FOR THE NUMBER OF CONTAINERS TO BE SHIPPED AND POSITION THE CONTAINERS ON THE TRAILER ACCORDINGLY.

TABLE I

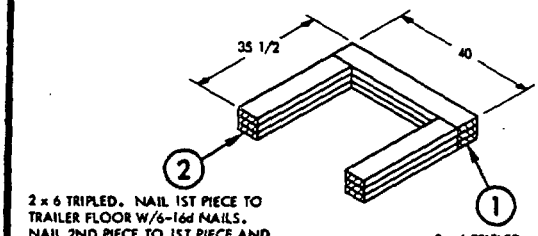
NO. OF CONTAINERS	LOAD PATTERN*	NO. OF CONTAINERS	LOAD PATTERN*	NO. OF CONTAINERS	LOAD PATTERN*
5		3		1	
4		2			

FWD

* LOAD PLAN SHOW FLOOR PLAN FOR TRAILER. ARROWS INDICATE CLAMP ASSEMBLY END OF CONTAINER.

2. WHEN SHIPPING FIVE BOOSTERS BLOCK AND TIE-DOWN USING THE PRINCIPLES OF THE FTL SHOWN ON PAGES 2 AND 3. LOCATE REAR CONTAINER AND THE NECESSARY PREPOSITIONED BLOCKING ON TRAILER'S CENTER LINE. USE A BACKUP CLEAT FOR THE OUTBOARD REAR CORNERS OF SECOND STACK (SEE PIECE 4, PAGE 5).
3. WHEN SHIPPING FOUR OR LESS BOOSTERS BLOCK AND TIE-DOWN USING THE PRINCIPLE OF THE LTL SHOWN ON PAGE 5.
4. A STACK OF ONE CONTAINER SHALL HAVE TWO TIE-DOWNS AND A STACK OF TWO CONTAINERS SHALL HAVE THREE TIE-DOWNS.

Source: https://assist.dla.mil -- Downloaded: 20261O-23T23:43Z
Check the source to verify that this is the current version before use.



2 x 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.

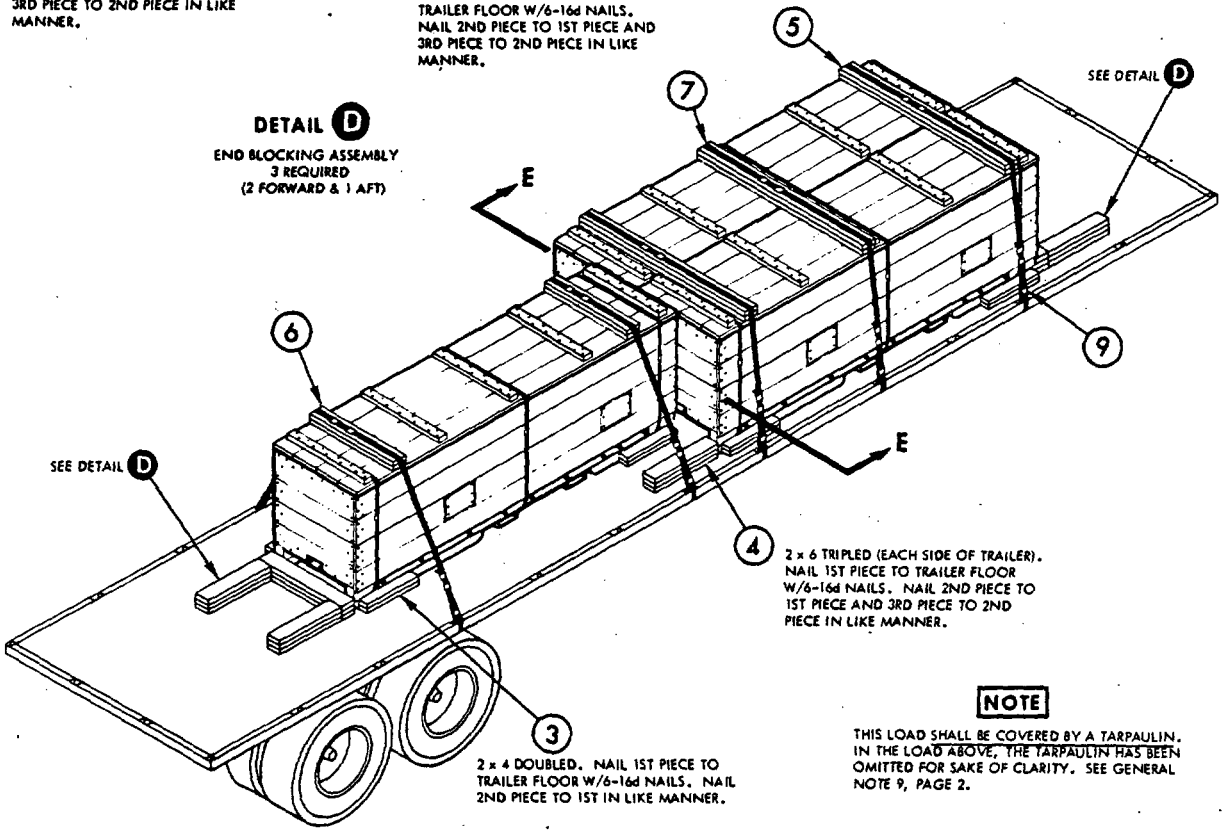
2 x 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.

NOTE

THE LTL ILLUSTRATES THE USE OF 2 x .050 STRAPS AS TIEDOWNS± CHAIN AND LOAD BINDERS ARE A AUTHORIZED ALTERNATE. SEE GENERAL NOTES 4, 6, 7 AND 8, PAGE 2.

DETAIL D

END BLOCKING ASSEMBLY
3 REQUIRED
(2 FORWARD & 1 AFT)

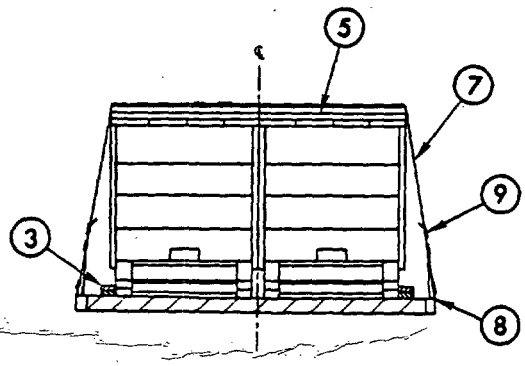


2 x 6 TRIPLED (EACH SIDE OF TRAILER). NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.

2 x 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

NOTE

THIS LOAD SHALL BE COVERED BY A TARPAULIN. IN THE LOAD ABOVE, THE TARPAULIN HAS BEEN OMITTED FOR SAKE OF CLARITY. SEE GENERAL NOTE 9, PAGE 2.



SECTION E-E

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE
9	SEAL	2 INCH	40	-	-	-
8	PAD	2 x .050 x 18	10	-	-	-
7	STEEL STRAP	2 x .050 TO SUIT	5	-	-	-
6	PROTECTOR BOARD	2 x 6 x 40	4	DOUBLE LAMINATE	4	8d
5	PROTECTOR BOARD	2 x 6 x 80	6	DOUBLE LAMINATE	8	8d
4	BACKUP CLEAT	2 x 6 x 30	4	SEE FIELD NOTE		
3	SLEEPER	2 x 4 x 24	16	SEE FIELD NOTE		
2	BACKUP	2 x 6 x 30	18	SEE DETAIL D		
1	CROSSMEMBER	2 x 6 x 40	9	SEE DETAIL D		

LIST OF MATERIALS AND NAILING DATA

LTL 40 FT TRAILER (FLATBED)

REVIEW ACTIVITY:
NAVY - OS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N291)

☆ U. S. GOVERNMENT PRINTING OFFICE: 1975-603-766/5003

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:43Z
Check the source to verify that this is the current version before use.